Languages: English, Spanish, Danish
Nationality: Danish, Norwegian, Cuban

Einar Benjamin Jensen
MSc in Autonomous Systems

Phone: +45 42432850
Email: ebjensen03@gmail.com
LinkedIn: einarj17

PROFILE

A hardworking and dedicated Mechatronics engineer with a strong passion for robotics, control systems, and automation technologies. Highly skilled in integrating electronics, programming, and mechanical design to develop intelligent robotic systems. Hands-on experience with medical technologies and a clear ambition to build a career within robotics and AI.

SKILLS

Programming	Python (pandas,NumPy), C, C++, Git, Object Oriented Programming
Robotics and Control	Linux/Ubuntu, ROS2 (rclcpp, rclpy, MoveIt2, ros2_control), MATLAB/Simulink, Adaptive Nonlinear Control, System Identification, State-Space Modelling, Motion Planning
Machine Learning	Reinforcement Learning, OpenCV, OpenPose, PyTorch, TensorFlow, Gymnasium, Stable Baselines3, PyBullet
Embedded Systems & DSP	Embedded Programming, Digital Signal Processing, Microcontroller Integration, Sensor interfacing, Sensor Fusion
CAD	Siemens NX, Creo Parametric, Autodesk Inventor, SolidWorks
Simulation	CFD Analysis in Star-CCM+, FEM Analysis in Ansys Workbench
Electrical	KiCad, LTSpice, PCB Prototyping, Analog Circuit Design
Soft Skills	Analytical problem-solving, strong collaborator, multidisciplinary teamwork

EDUCATION

Technical University of Denmark (DTU) — Sep 2026- Jan 2028
MSc Honours Programme in Autonomous Systems

University of Southern Denmark (SDU) — Sep 2022- Jan 2026
BSc Mechatronics Engineering — Honors 10.3/12 GPA

National University of Singapore (NUS) — Aug 2024-Dec 2024
Exchange Programme, Robotics and Automation

EXPERIENCE

Bachelor Thesis Writer — Aug 2025-Jan 2026
Novo Nordisk, — Received Grade: 12/12
Copenhagen, Denmark

Thesis title: Modelling and Control of a Time-Pressure Membrane Valve System for Pharmaceutical Filling

- Contributed to a novel filling line concept through a grey-box modelling approach combining physics and data-driven methods.
- Designed an embedded system architecture for real-time sensor data acquisition and model validation
- Built Python data pipelines to process high-fidelity CFD simulation outputs for model training and validation
- Developed and validated nonlinear control strategies compared to baseline ON/OFF control.

Manufacturing Automation Technology Intern Feb 2025-July 2025
Danfoss Drives, Full-time
Gråsten, Denmark

- Designed and deployed a robot assembly cell for human-robot collaboration complying with ISO standards
- Programmed serial-chain manipulators for kitting operations, involving inter-robot communication.
- Tested and redesigned frequency converter parts in CAD to optimise assembly efficiency per DFAA principles.
- Developed and evaluated vision-based classification systems for component recognition, comparing commercial solutions with custom deep learning models using OpenCV.

PROJECTS

Reinforcement Learning for Serial-Chain Manipulator Control April 2026

- Built a custom OpenAI Gym environment for a 7-DOF Kuka iiwa arm in PyBullet
- Utilized a 17-dimensional observation space and a negative Euclidean distance reward function to guide policy learning
- Trained an on-policy PPO (Proximal Policy Optimization) agent using Stable Baselines3 with a custom MLP policy network [512,512,256,128] and ReLU activation achieving end effector positioning within 2cm of random fixed targets
- Two-stage training conducted using an additional reward term for angular velocity penalization to reduce joint oscillation near target

Deep Learning for Pose and Age Classification Apr 2025-Jun 2025

- Implemented an OpenPose-based python classifier with skeleton overlay for automated posture recognition in Jupyter Notebook
- Processed and structured live video data streams in Python using OpenCV and NumPy, applying real-time inference and logging prediction outputs for model evaluation

Robot Manipulator Path Planning using STOMP Sep 2024-Nov 2024

- Developed a stochastic trajectory optimisation algorithm in MATLAB for collision-free motion planning of serial-chain manipulators
- Integrated forward kinematics using the Product of Exponentials formulation
- Maintained end-effector orientation constraints whilst simulating obstacle avoidance

Quadcopter Stabilisation Feb 2024-Jun 2024
Semester Project 4

- Modelled quadcopter dynamics in state-space representation
- Developed an LQI controller in MATLAB and Simulink and deployed it in BeagleBone Blue
- Collected motor data on force-thrust characteristics for control system analysis and model validation
- Developed a Kalman filter for sensor fusion and state estimation, reducing process and measurement noise
- Designed and 3D printed a drone frame mechanically validated in Ansys

ROS2 Vision-Guided Picking - Kuka iiwa 7-DoF May 2026

- Built a complete vision-to-motion pipeline in ROS2 Humble
- Python vision node publishes detected object poses on a ROS2 topic, a C++ MoveIt 2 client subscribes, runs inverse kinematics via the Pilz PTP planner, and executes the trajectory on a simulated Kuka iiwa arm
- Authored C++ subscriber, publisher, and MoveIt 2 client nodes using rclcpp; authored Python publisher node using rclpy
- Configured ros2_control with JointTrajectoryController at 225Hz; visualised live joint telemetry with PlotJuggler